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Unit 3 - Designing Solutions

PDF Documents

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UNIT 3 DESIGNING SOLUTIONS

Programming Concepts

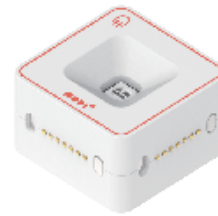
WHILE LOOP

A "while" loop continues to execute its block of code **as long as the condition is true**. It checks the condition before running the loop's code. If the condition is true, the loop runs. If it's false, the loop stops. For example, "while the sun is up, keep the LED off."

✓ WHILE



condition
While the sun is up,



task
Keep the LED off

Another Example:

"Think of using a 'while' loop like brushing your teeth while a timer is running. You start brushing, and 'while' the timer hasn't beeped (meaning time is not up yet), you keep brushing. As soon as the timer beeps, signaling the time is up, you stop brushing."



condition
While the timer is running,



task
Keep brushing your teeth.



Until statement

In programming, the "until" statement is like having a goal and doing something over and over until you reach that goal.

✓ **UNTIL**

\$1000



condition

Until you reach
your goal of \$1,000,

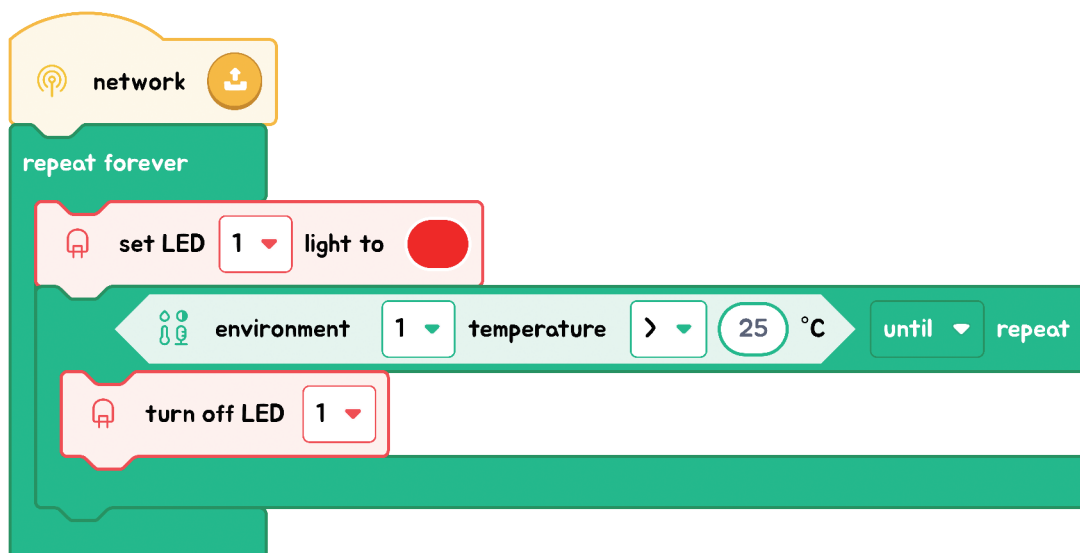
task 1

Keep saving money.

For example, if your goal is to save \$1,000, you keep adding money to your savings "until" you have \$1,000. Just like saving money bit by bit until you hit your target, the "until" statement repeats an action until a certain condition is met.

Another Example:

In our example, think of the LED as a light that stays off "until" it gets warm enough, say 25 degrees. So, the LED doesn't turn on while it's cooler.



But once it hits 25 degrees or more, the red light comes on. The "until" statement here waits for the temperature to go up, and when it does, it's like a signal to turn the red light on.

Flowcharts & Algorithms

Let's read the flowchart and create the code.

FLOWCHART

